

ME2 Computing- Coursework summary

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Path: A

Words: /1000

A) What physics are you trying to model and analyse? (Describe clearly, in words, what physical phenomenon you wish to analyse)

The physics we aim to model and analyse is the propagation of sound waves, which are mechanical waves, in a confined space. In this case, we will be modelling the propagation of an instantaneous single-pitch sound generated from a point source, which may be for example Nicolas' voice at the start of his lectures. This voice generates a sound wave that travels through the air and reflects off the walls of CAGB 203, interacting with the environment. Using the 2D wave equation, which describes how sound waves move through a medium, in this case, air, within the boundaries of a two-dimensional space, one can study how these sound waves behave, how they reflect off the walls, and how they spread throughout the room, helping us to understand various acoustic phenomena, such as echoes, reverberation, and the distribution of sound energy within the room. Assuming and modelling the lecture room as a square room, and Nicolas' voice as a brief 'note': a single-pitch point source, we can then set Dirichlet boundary conditions to zero at the walls to model this phenomenon (each specific boundary condition will be explained in detail later).

The initial sound wave "produced" is modelled by a function $f(x,y)$, which defines the amplitude, the position, and the shape of the wave desired.

B) What PDE are you trying to solve, associated with the Physics described in A? (write the PDE)

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (1)$$

This is a hyperbolic PDE in 2D. 2nd order in space, and 2nd order in time.

C) Boundary value and/or initial values for my specific problem: (be CONSISTENT with what you wrote in A)

The following conditions are set:

$$u_{[0,x,y]} = f(x,y) = e^{(-350*((x-1)^2+(y-1)^2))} \quad (2)$$

The initial condition in equation 2 represents the initial shape of the wave at time = 0, i.e. the initial distribution of sound in the room in the immediate moment that Nicolas starts speaking. It is modelled as a Gaussian that arises naturally in many physical phenomena of sound attenuation: the Gaussian distribution peaks at its centre and decays smoothly towards zero as distance increases from the centre. This represents the stated scenario, where the sound is emitted from a localized source (Nicolas) and gradually dissipates as it spreads out in all directions. This initial distribution is symmetric and smooth, making it mathematically tractable and computationally simple. The function is shaped at the centre [1,1] which is modelled as Nicolas' position in the lecture room.

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = 0 \quad (3)$$

Equation 3 shows the initial Neumann condition representing the starting velocity, i.e. the change of the wave displacement at time =0. In theory, when someone speaks, the motion of their vocal cords implies a non-zero starting velocity: it causes the air particles in the near surroundings to move/vibrate. This magnitude of movement, however, is deemed many orders of magnitude smaller than the speed of sound propagation through air. In scenarios involving turbulent flow or high-frequency sound waves, accounting for the initial velocity of air particles is crucial for accurate results, especially in environments like wind tunnels or jet engines where motion dynamics are significant. However, in room acoustics and speech modelling, where the emphasis is on lower-frequency sound waves and longer time scales, the initial velocity of sound emitted by a speaker is typically negligible compared to the speed of sound in air. Therefore, setting the initial velocity set to zero is a simplification that balances computational efficiency with accuracy for the application at hand.

$$u_{[0,y,t]} = 0 \quad (4)$$

$$u_{[1,y,t]} = 0 \quad (5)$$

$$u_{[x,0,t]} = 0 \quad (6)$$

$$u_{[x,1,t]} = 0 \quad (7)$$

The remaining boundary conditions described by equations 4 to 7 show zero displacement at the boundaries i.e. at the walls. This follows from the assumption that the walls of the lecture hall are perfectly reflecting boundaries, meaning that the wall does not transmit any noise through it, nor does it absorb it in the form of heat: the incident wave is instead fully and perfectly reflected, without any attenuation or phase change. This is a reasonable assumption to make, especially in the context of the lecture hall where the most significant phenomenon will be echo, especially if the lecture hall is empty. It is also worth noting that the derivative at the walls is in fact non-zero.

D) What numerical method are you going to deploy and why? (Describe, in words, which method you intend to apply and why you have chosen it as opposed to other alternatives)

The equation will be numerically solved by applying finite differences to the partial derivatives. More precisely, central finite difference is applied to the three partial derivatives. This simplification will establish a relationship between a point and the values of multiple points that influence its value.

Next, an explicit method will be used to isolate the point in "the future" (i.e., the point that needs to be determined) from the other points at previous time steps.

An explicit method has been preferred over an implicit method after evaluating the main characteristics of the two:

EXPLICIT	IMPLICIT
Algorithm less complex	More complex algorithm
Convergence and stability aren't guaranteed	Convergence is unconditional
Not suitable for stiff problems (time step required would be very small)	The method is 1 st order accurate and loses accuracy when time step increases

D) What numerical method are you going to deploy and why? (Contd...) As presented in Table 1, the main advantage of the implicit method over the explicit is its unconditional convergence. This however comes at the cost of computational efficiency; implicit methods utilise a more complex algorithm and require more computational power. If a suitable convergence criterion is followed, explicit methods may prove more advantageous. In this case, our problem is not 'stiff' - it does not require very small time steps for convergence (which can be computationally expensive) but instead the equation in use is simple enough to not pose stiffness issues, rendering the explicit method more adequate.

E) I am going to discretise my PDE as the following (show the steps from continuous to discrete equation and boundary/initial conditions:

We start from the continuous 2D wave equation (equation 1):

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (8)$$

Where u is a scalar function of time and space $u(t, x, y)$ and represents the displacement of the wave from its equilibrium position (in this case, since our wave is a sound wave, it represents the variation in air pressure caused by the sound wave; t is time, x and y are positions in space and c is the wave speed, in this case the speed of sound in the chosen medium, air (≈ 343 m/s), which is assumed to be constant.

Firstly, we introduce a mesh in time and in space. The mesh in time consists of time points:

$$t = 0 < t_1 < \dots < t_N \quad (9)$$

with a constant time spacing $\Delta t = t_{n+1} - t_n = k$, and the mesh in space:

$$x_0 < x_1 < \dots < x_{N_x} \quad (\text{in the } x \text{ direction}) \quad (10)$$

$$y_0 < y_1 < \dots < y_{N_y} \quad (\text{in the } y \text{ direction}) \quad (11)$$

where a constant mesh spacing is also used: $\Delta x = x_{i+1} - x_i$, and since a square domain is assumed and for simplicity, $\Delta x = \Delta y = h$.

The unknown u at a mesh point t_n, x_i, y_j is denoted by $u_{i,j}^n$.

Before proceeding with the discretization, the equation is non-dimensionalised; this allows to simplify the problem and facilitates scaling analysis, whilst also allowing for the identification of crucial dimensionless parameters (if any), reducing computational effort and improving the interpretation of results. The dimensionless form of the wave equation is as shown below (2):

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \quad (6)_1$$

Where both x and y now range from $0 \leq x \leq 1$ and $0 \leq y \leq 1$.

The difference quotients corresponding to the partial derivatives are then approximated using central finite difference in both time and space (equations 3 and 4):

$$\begin{aligned}\frac{\partial^2 u}{\partial t^2} &\approx \frac{1}{k^2} [u(t+k, x, y) - 2u(t, x, y) + u(t-k, x, y)] \\ \frac{\partial^2 u}{\partial x^2} &\approx \frac{1}{h^2} [u(t, x+h, y) - 2u(t, x, y) + u(t, x-h, y)] \\ \frac{\partial^2 u}{\partial y^2} &\approx \frac{1}{h^2} [u(t, x, y+h) - 2u(t, x, y) + u(t, x, y-h)]\end{aligned}\tag{7}$$

Central difference was preferred because it is more stable, accurate and has a faster convergence rate; this is significant since an explicit method is being employed, meaning that stability is of importance and hence central difference is more adequate.

Substituting equations 3,4 and 5 into equation 2 and writing in index notation form leads to equation 6:

$$\frac{1}{k^2} [u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1}] = \frac{1}{h^2} [u_{i+1,j}^n + u_{i-1,j}^n + u_{i,j+1}^n + u_{i,j-1}^n - 4u_{i,j}^n]\tag{9}$$

Letting $r = \frac{k^2}{h^2}$ and rearranging equation 6 becomes:

$$[u_{i,j}^{n+1} + u_{i,j}^{n-1}] = r[u_{i+1,j}^n + u_{i-1,j}^n + u_{i,j+1}^n + u_{i,j-1}^n] + (-4r + 2)u_{i,j}^n\tag{10}$$

This equation is insightful when determining the condition for stability; since the contribution of the $u_{i,j}^n$ term must be positive, one can establish that $-4r + 2 \geq 0$, hence $r \geq \frac{1}{2}$. Taking the limiting case $r = \frac{1}{2}$, the final discretised equation becomes:

$$u_{i,j}^{n+1} = \frac{1}{2} (u_{i+1,j}^n + u_{i-1,j}^n + u_{i,j+1}^n + u_{i,j-1}^n) - u_{i,j}^{n-1}\tag{8}$$

For the initial condition at $t = 0$, an issue arises with the final $u_{i,j}^{n-1}$ term. This value lies outside the mesh at a negative time $t = -k$; such 'ghost value' can be determined from the central finite difference of the Neumann boundary condition established as $\frac{\partial u}{\partial t}\bigg|_{t=0} = 0$:

$$\frac{\partial u}{\partial t} \approx \frac{1}{2k} (u_{i,j}^{n+1} - u_{i,j}^{n-1}) = 0$$

which gives that $u_{i,j}^{n-1} = u_{i,j}^{n+1}$. A special case equation (10) can hence be derived for $t = 0$:

$$u_{i,j}^1 = \frac{1}{4} (u_{i+1,j}^0 + u_{i-1,j}^0 + u_{i,j+1}^0 + u_{i,j-1}^0)$$

Equation 8 can then be used for the remaining $t > 0$ terms. The initial scenario is now fully defined: all the terms at $t = 0$ can be determined from the initial and boundary conditions, and it is then possible to iterate up until the desired time duration.

F) Plot the numerical results comprehensively and discuss them (discuss how the results describe the physics and comment on any discrepancies or unexpected behaviours). Use multiple types of visual graphs. Present and discuss any outcomes of the grid analysis, as requested in Task 9, too.

The wave develops in the room as per [this animation](#). Snapshots of the wave propagation at different times taken from the animation are depicted in Figures 1 to 4;. This is also plotted as [a contour animation](#) (see link for animation) developing over time.

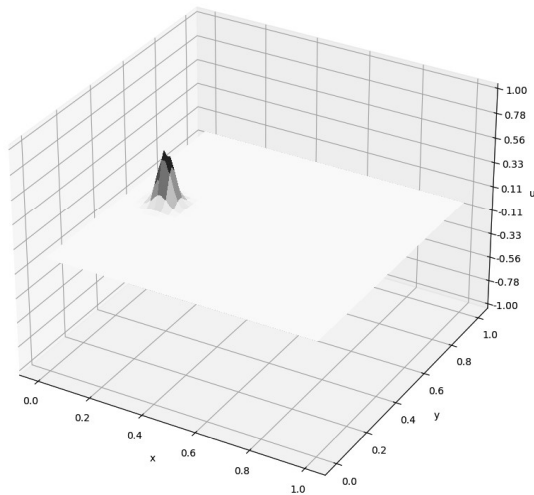


Figure 1 - acoustic wave source at $t=0s$

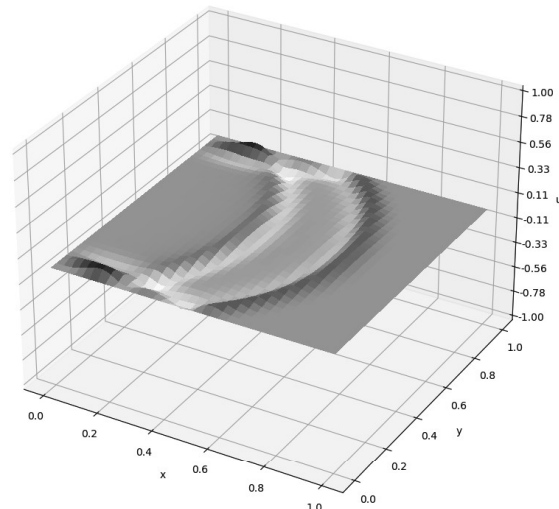


Figure 2 - acoustic wave at $t = 0.06s$

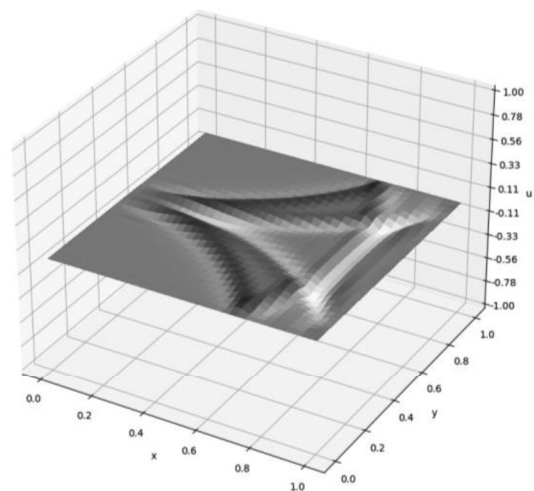


Figure 3 - acoustic wave at $t=0.1s$

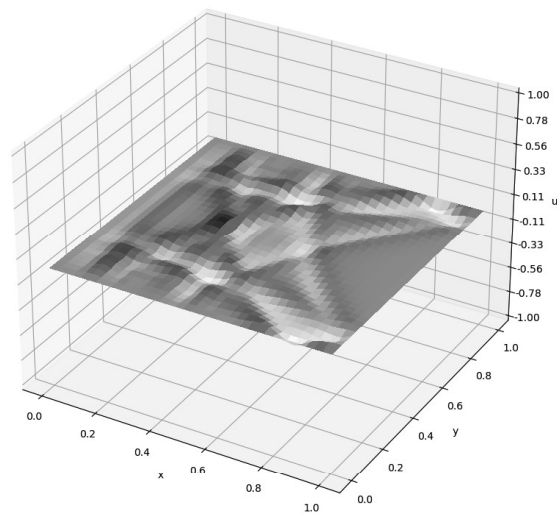


Figure 4 - acoustic wave at $t=0.18s$

The source contour and two development contours are also presented in Figure 5 to 7 (page 5).

F) Plot the numerical results comprehensively and discuss them (discuss how the results describe the physics and comment on any discrepancies or unexpected behaviours). Use multiple types of visual graphs. Present and discuss any outcomes of the grid analysis, as requested in Task 9, too. (Contd...)

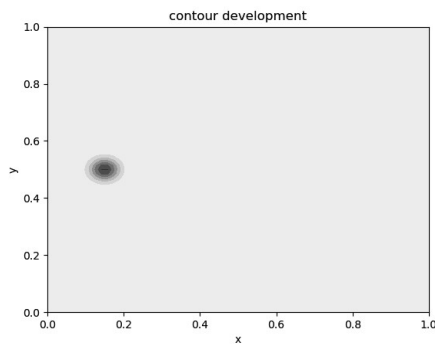


Figure 5 - contour source (t=0s)

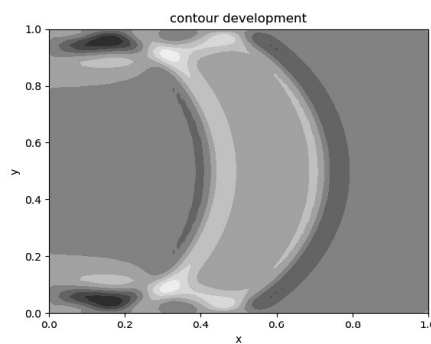


Figure 6 - contour at t=0.06s

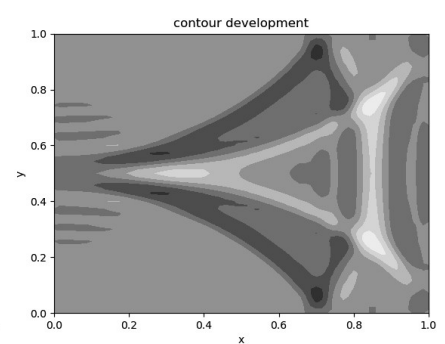


Figure 7 - contour at t=0.1s

It could be useful to understand how a specific plane is affected by the wave: this can be done by “slicing” the space at a specific point. Here, the vertical plane parallel to the x axis and passing through the source point was chosen and Figure 8 to 10 present three instants along this plane.

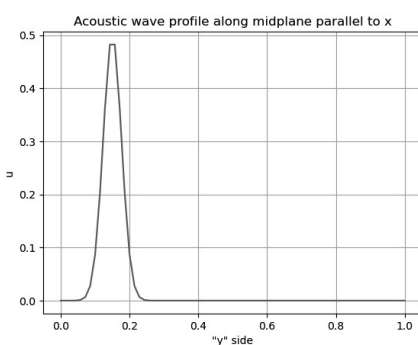


Figure 8 - t=0s

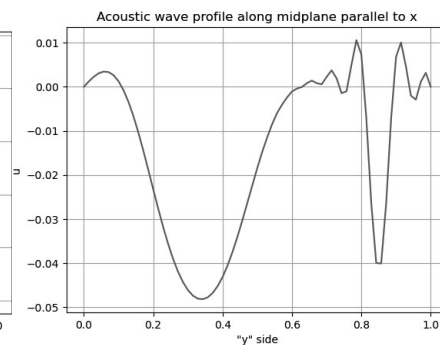


Figure 9 - t=0.1s

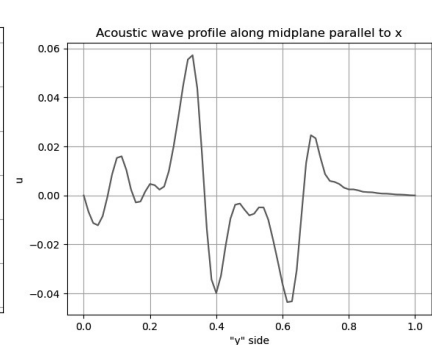


Figure 10 - t=0.18s

The plots reflect the expected behaviour of the physics studied. The sound wave is modelled as an instantaneous emission as seen in Figure 1 and 8, showing a single peak of maximum amplitude at the source position at time = 0, with the surroundings assumed to be quiet. After certain time interval, the wave is reflected from the walls and new profiles form. This follows from the wave's nature and the principle of wave superposition: when two waves interfere with each other, they may interact in a constructive or destructive manner. The first case occurs when two crests of the same sign combine, whilst the second occurs when two opposite-signed crests interfere. This interaction leads to the generation of amplified peaks or neutralized “flat areas”. Both phenomena can be observed in figures 3 and 7 (the plot and the contour), where the darker area, symmetrical about the midplane passing through the source, shows the constructive effect. In the same two figures a lighter/white area may also be identified, depicting the destructive effect, where peaks have ‘neutralized’.

The wave propagation behaviour seen in the animations also follows the expected propagation shape before and after reaching the walls (i.e. the obstacles) and getting reflected.

F) Plot the numerical results comprehensively and discuss them (discuss how the results describe the physics and comment on any discrepancies or unexpected behaviours). Use multiple types of visual graphs. Present and discuss any outcomes of the grid analysis, as requested in Task 9, too. (Contd...)

The first grid size chosen is $h=0.01429$, which results in 70 intervals on the x and y axis. The convergence criterion allows us to determine the maximum time step possible for this method to converge which results to be $k_{\max}=0.0101$. The time step chosen for this analysis is $k=0.001$ which ensures convergence.

The time elapsed to compute the numerical method successfully is recorded and equal to $T = 8.91$ s.

The results of a 10x mesh size reduction are presented in Figure 11 below. The effect of a finer/smaller step in the space domain is noticeable in the smoother surfaces and curves generated. This decrease in mesh size however leads to an increased time for computation, where the time elapsed to compute the numerical method for all the grid points is increased by a factor of 130x, rendering the computation too slow. In addition, maximum time step size for convergence was previously identified as $k_{\max}=0.00101$, whilst the time step in use here is $k=0.001$, which lies very to the limit of convergence. This grid size hence highlights a limitation of the explicit method that will be discussed later in Section G.

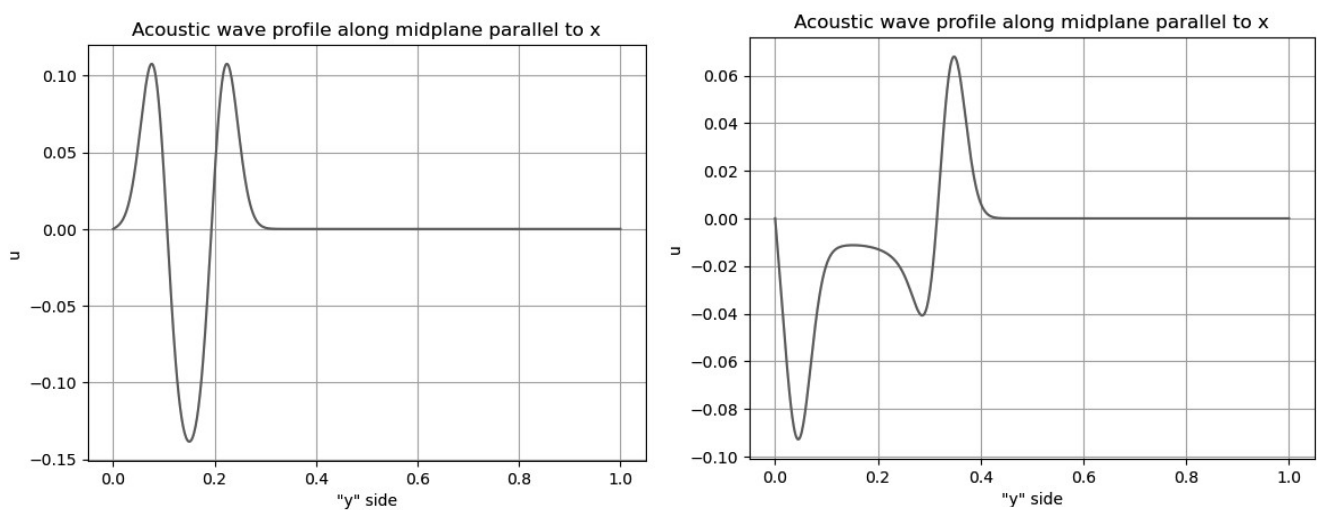


Figure 11: wave generated with 10x mesh size reduction in space at different time snapshots.

If an even greater grid reduction of 50x is utilised, the maximum time step now becomes of $k_{\max}=0.0002$. If the utilised time step remains 0.001, the method does not converge and this case is again further discussed in Section G.

G) Other remarks (limits of the model, convergence problems, possible alternative approaches, anything you find relevant and important to mention):

In essence, the model performs optimally when the mesh size is relatively large. Once the step in space, h , is reduced (taking as an example the 10x reduction case in Section F), the computational time increases exponentially. It is hence worth noting that time is the main limitation of this method when operating within the convergence regime but with a decreased mesh size.

The other main limitation was already determined initially: the time step is bounded by the convergence criterion – defined during the method evaluation – to a specific value depending on the mesh size. This means that a relatively small mesh size is hence required to mitigate the problem of convergence for a fixed time step of 0.001s. If the grid is reduced to be close to the divergence limit the main issue hence becomes the waiting time for the computation of the result.